# The first Hochschild cohomology group of Brauer graph algebras with multiplicity one

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## Schroll 2015

Brauer graph algebras with multiplicity function identically one.

$$\Updownarrow$$

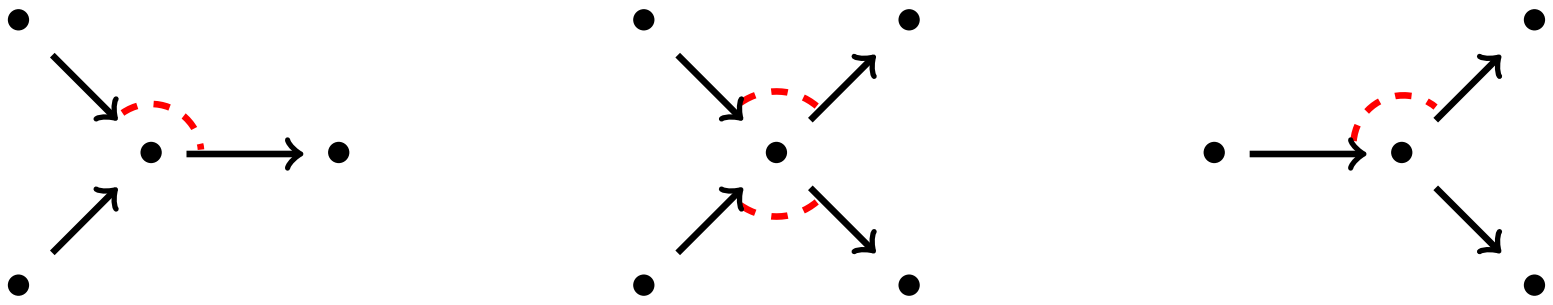
Trivial extension of gentle algebra.

# Preliminaries

## Definition

A finite dimensional algebra $A$ is gentle if it is Morita equivalent to an algebra $kQ/I$ satisfying

1. Each vertex of $Q$ is the starting point of at most two arrows and is the end point of at most two arrows.
2. For each arrow $\alpha$ in $Q$ there is at most one arrow $\beta$ in $Q$ such that $\beta\alpha \notin I$ and there is at most one arrow $\gamma$ such that $\alpha\gamma \notin I$.
3. For each arrow $\alpha$ in $Q$ there is at most one arrow $\beta$ in $Q$ such that $\beta\alpha \in I$ and there is at most one arrow $\gamma$ such that $\alpha\gamma \in I$.
4. $I$ is generated by paths of length $2$.



Given an algebra $A$ we consider the dual $A$-bimodule of $A$

$$DA = Hom_k(A, k)$$

with structure given by $bfa(x) = f(axb)$ for $a, b \in A$ and $f \in DA$.

## Definition

The trivial extension $TA$ is the algebra whose underlying vector space is $A \oplus DA$ with the multiplication defined by

$$(a, f)(b, g) = (ab, ag + fb)$$

for $a, b \in A$ and $f, g \in DA$.

# A decomposition theorem

Cibils, Marcos, Redondo and Solotar (2003) show that if $A$ is a finite dimensional algebra, the first Hochschild cohomology group $HH^1(TA)$ of the trivial extension of $A$ is the sum of four vector spaces,

$$HH^1(TA) = Z(A) \oplus HH^1(A) \oplus HH_1(A)^* \oplus Alt_A(DA),$$

where $Z(A)$ is the centre of $A$ and $Alt_A(DA)$ is a set of skew-symmetric bilinear forms given by

$$Alt_A(DA) = \{\varphi \in Hom_{A-A}(DA, A) : f\varphi(g) + \varphi(f)g = 0\}.$$

We will use the Cibils' ideas to compute $HH^1(TA)$.

# Notation

- $k$ is an algebraically closed field with $char(k) \neq 2$,
- $Q$ is a finite connected quiver,
- $A = kQ/I$ is a finite dimensional gentle $k$-algebra where $I$ is generated by a minimal set $R$ of paths of length 2,
- $B$ is the set of paths of Q which do not contain any path of $R$.

# $HH^*(A)$

We interpret the Hochschild cohomology spaces $HH^0(A)$ and $HH^1(A)$ in terms of parallel paths.

## Definition

Two paths $\varepsilon$, $\gamma \in Q$ are called parallel if $\varepsilon$ and $\gamma$ start at the same vertex, and they end at the same vertex.

$$\begin{tikzcd} e_i \arrow[r, bend left, "\varepsilon"] \arrow[r, bend right, "\gamma"'] & e_j \end{tikzcd}$$

If $X$ is a set of paths in $Q$,

$$X||B := \{(\varepsilon, \gamma) \in X \times B : \varepsilon \text{ and } \gamma \text{ are parallel}\}.$$

## Proposition (Cibils-Saorín, 2001)

*Let $A$ be a monomial algebra. The Hochschild cohomology of $A$ in low degrees is the cohomology of the cochain complex*

$$0 \longrightarrow k(Q_0||B) \xrightarrow{d^0} k(Q_1||B) \xrightarrow{d^1} k(R||B) \to ...$$

*where*

$$d^0(e,\gamma) = \sum_{a \in Q_1 e} (a, a\gamma) - \sum_{a \in eQ_1} (a, \gamma a),$$

$$d^1(a,\gamma) = \sum_{p \in R} (p, p^{(a,\gamma)}),$$

*with $p^{(a,\gamma)}$ the sum of all nonzero paths obtained by replacing one appearance of the arrow $a$ in $p$ by the path $\gamma$.*

# $HH^0(A)$

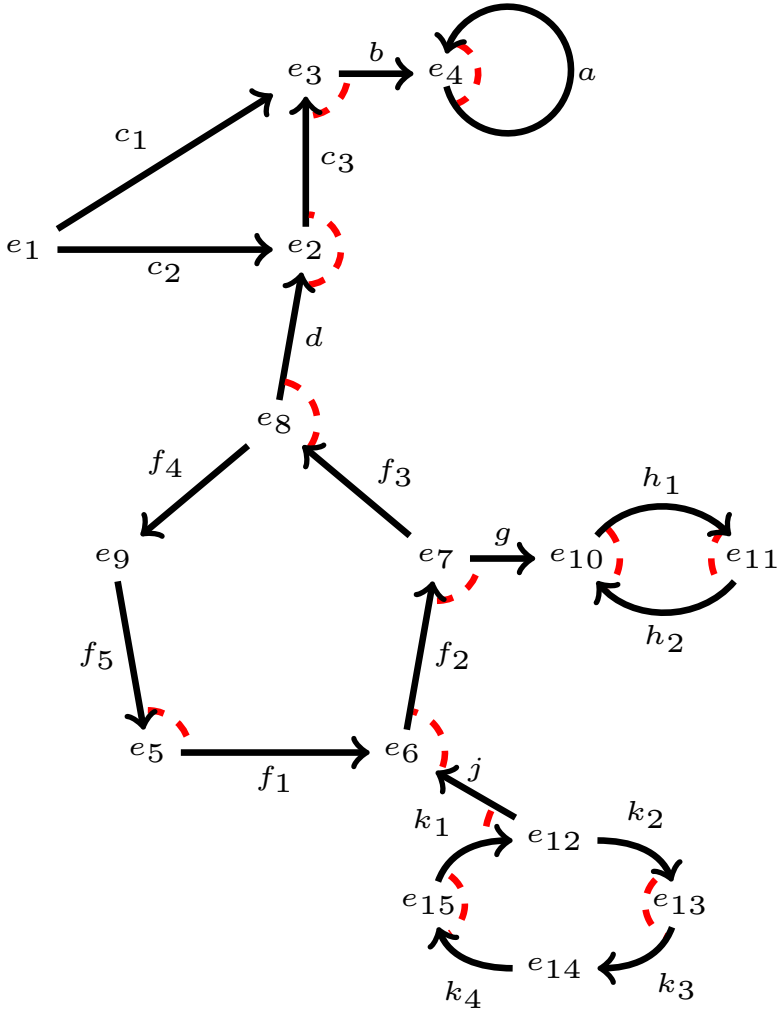
**Theorem**

*Let $A$ be a finite dimensional gentle algebra. The set of the following elements is a basis of $HH^0(A)$:*

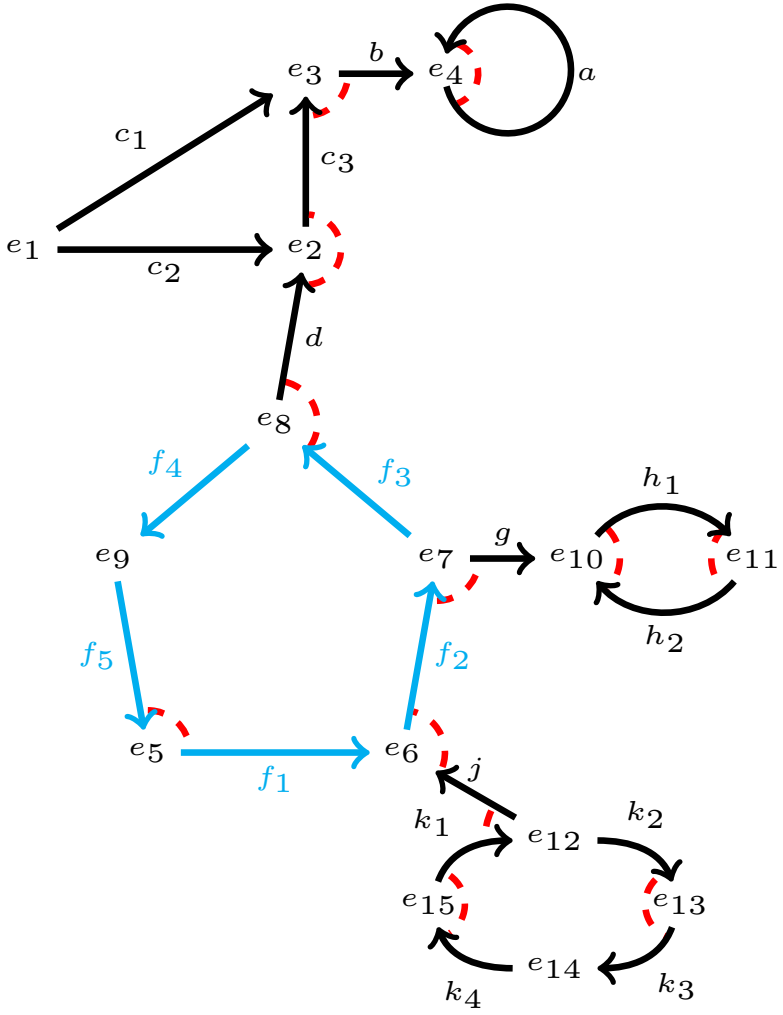
- $\sum_{e_i \in Q_0} (e_i, e_i)$,
- *The pairs* $(e, \gamma) \in Q_0 || B_n$ *where* $n \geq 1$ *and* $val(e) = 2$.

## Example

Let $A = kQ/I$ be the following gentle algebra



# Example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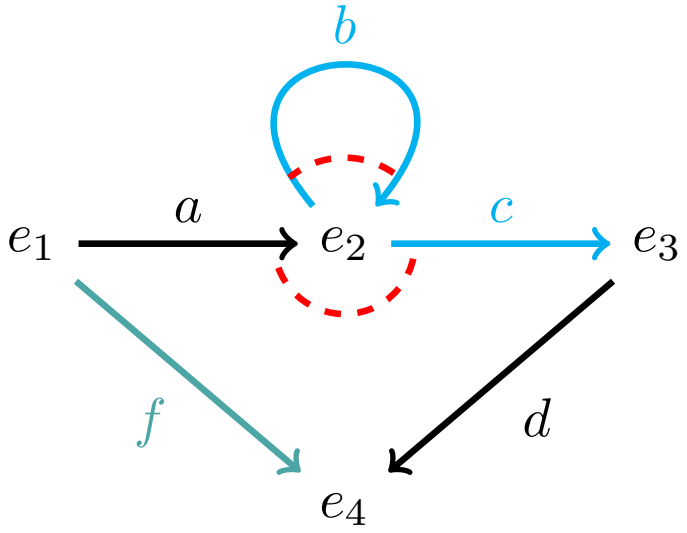
# $HH^1(A)$

## Definition

Let $(a, \gamma) \in Q_1||(B \setminus Q_0)$ with $a \neq \gamma$. If $a$ does not appear in $\gamma$ we say that $a$ is a shortcut for $\gamma$, on the contrary, we say that $\gamma$ is a deviation via $a$.

## Example

In the quiver we have $(f, dcba) \in Q_1||B_4$ and $f$ is a shortcut for $dcba$, whereas $(c, cb) \in Q_1||B_2$ and $cb$ is a deviation via $c$.



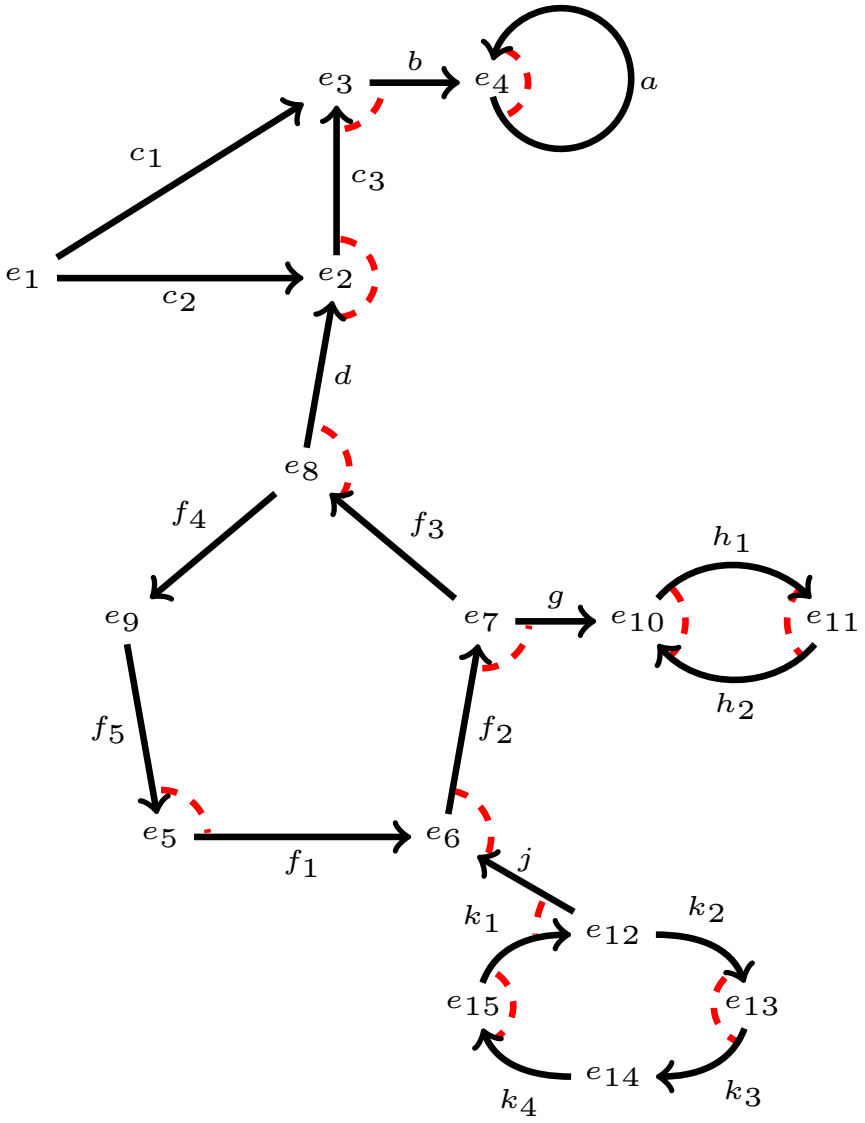
# $HH^1(A)$

## Theorem (Schroll-Solotar-C)

*Let $A$ be a finite dimensional gentle algebra. The following elements give a basis of $HH^1(A)$:*

- *The pairs $(a, \gamma)$ such that $a$ is a shortcut for $\gamma$ and $a$ is not in any relation in $R$,*
- *The pairs $(a, \gamma)$ where $\gamma$ is a deviation via $a$, $a$ is not in any relation in $R$ and $\gamma = \gamma_2 a \gamma_1$ with $\gamma_1, \gamma_2 \notin Q_0$,*
- *The pairs $(a, a)$ for each minimal simple cycle in $Q$ where $a$ is determined by $Q_1 \setminus T_1$, with $T$ a fixed maximal tree in $Q$.*

# Example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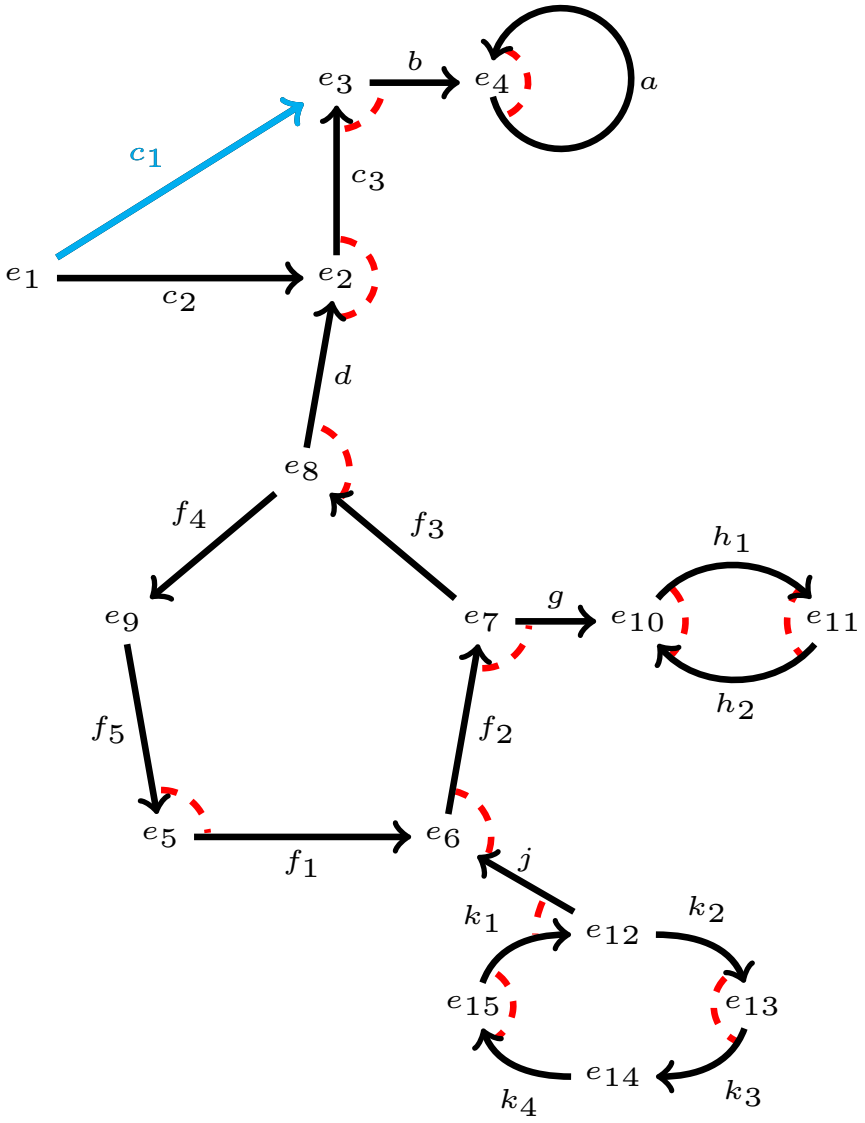
# Shortcuts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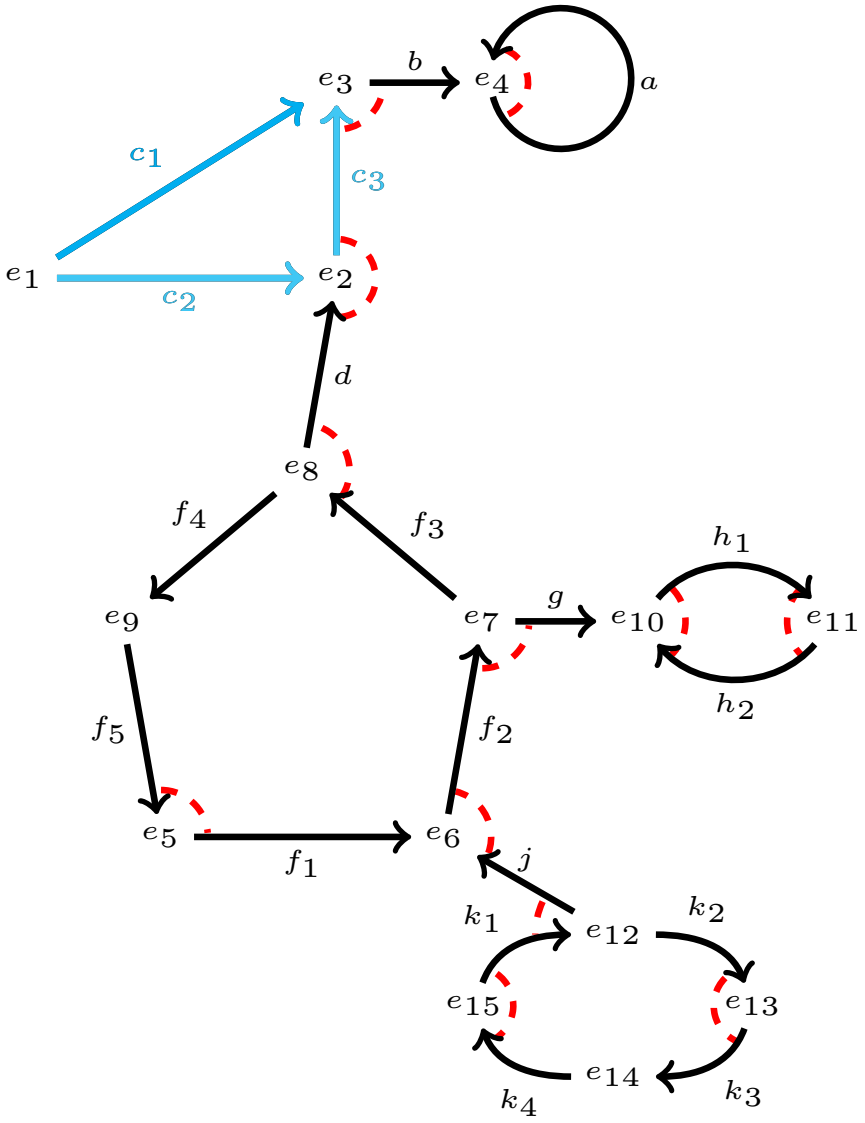
# Shortcuts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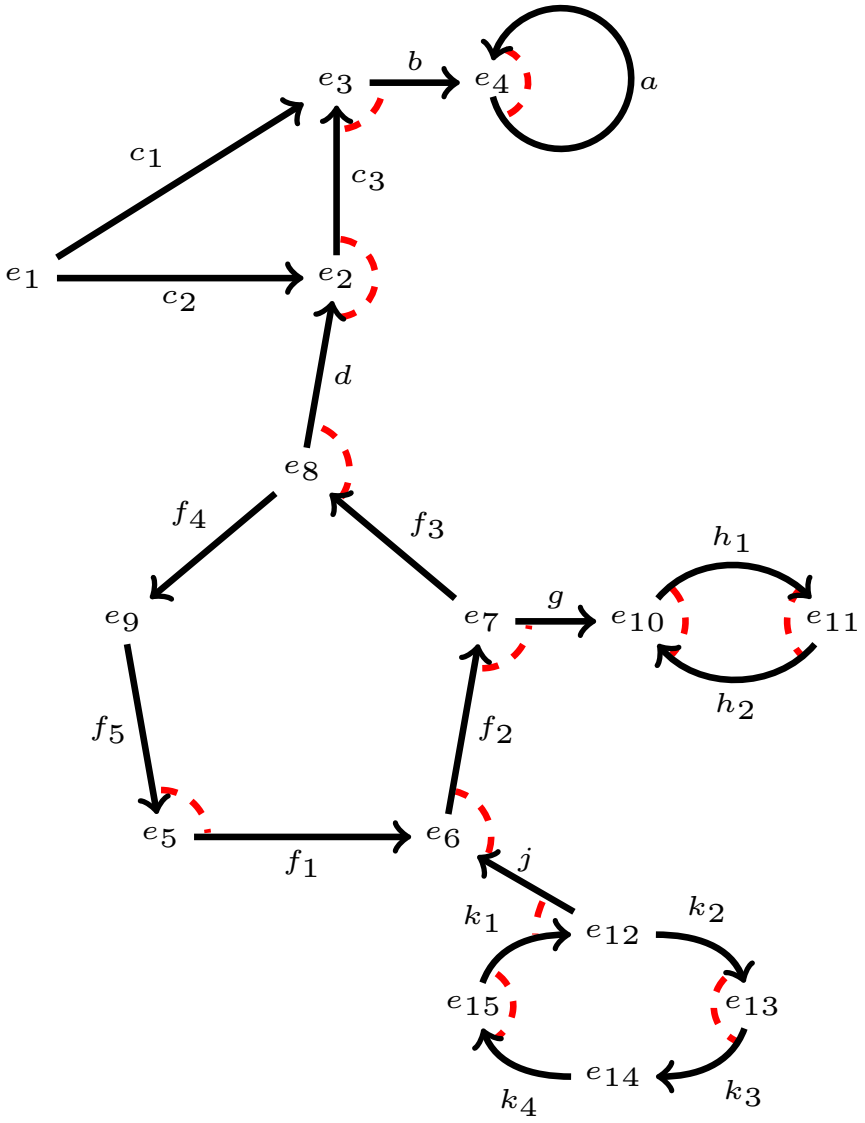
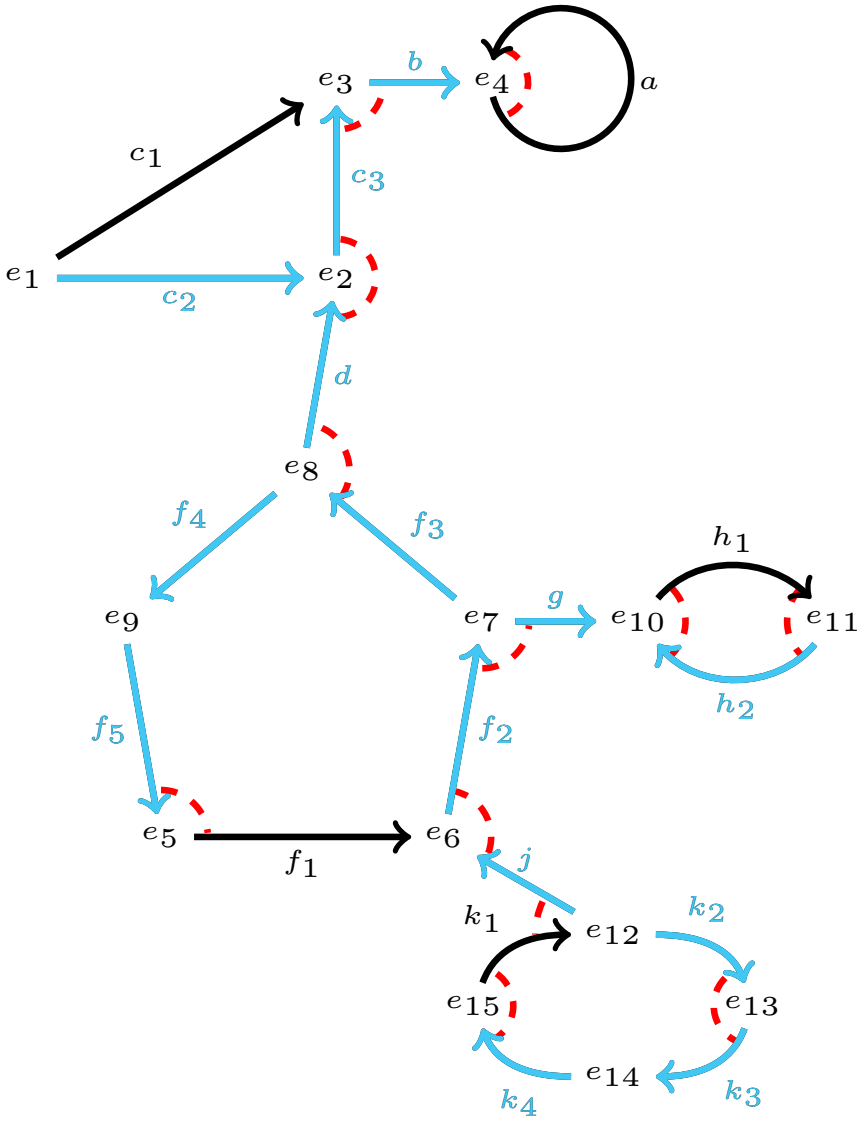
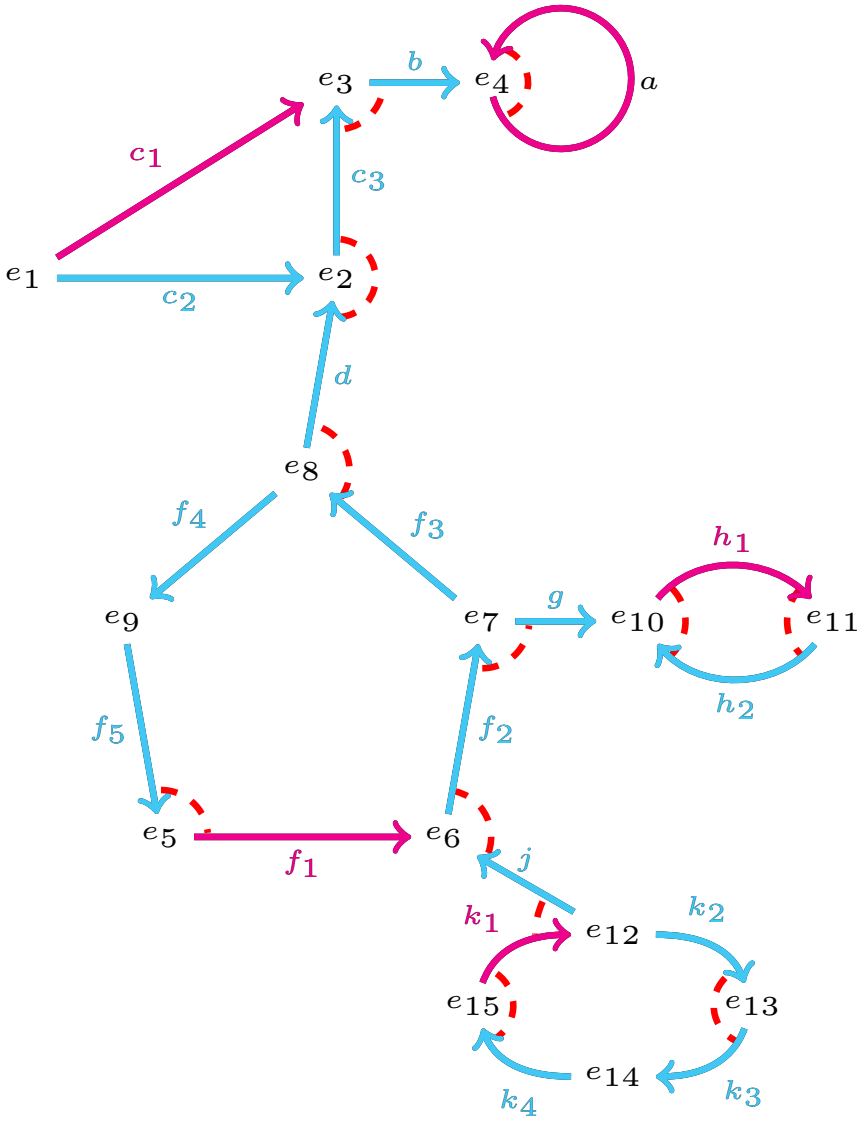
# Deviations: We do not have it ☹



# The tree $T$ and its complement $U$



# The tree $T$ and its complement $U$

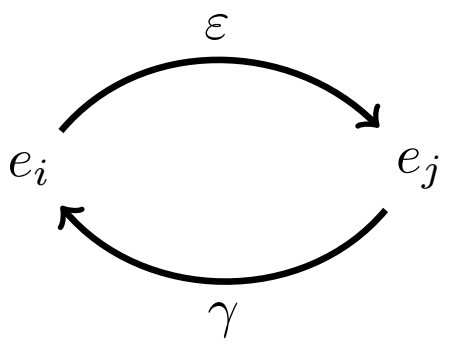


# $HH_1(A)^*$

We interpret the dual of the first Hochschild homology space $HH_1(A)^*$ in terms of cyclic pairs.

## Definition

Two paths $\varepsilon$, $\gamma \in Q$ are a cyclic pair if the vertex where $\varepsilon$ starts is the vertex where $\gamma$ ends and vice versa.




If $X$ is a set of paths in $Q$, we denote by

$$X \odot B^* := \{(\varepsilon, \gamma^*) : (\varepsilon, \gamma) \in X \times B, \ \varepsilon \text{ and } \gamma \text{ are a cyclic pair}\}.$$

$B^*$ is the basis of the dual vector space $DA$.

## Proposition (Strametz, 2002)

*Let $A$ be a finite dimensional monomial algebra. The dual of the Hochschild homology of A in low degrees is the cohomology of the cochain complex*

$$0 \longrightarrow k(Q_0 \odot B^*) \xrightarrow{d_0} k(Q_1 \odot B^*) \xrightarrow{d_1} k(R \odot B^*) \to \ldots$$

*where*

$$d_0(e, \gamma^*) = \sum_{a \in Q_1 e} (a, a\gamma^*) - \sum_{a \in e Q_1} (a, \gamma^* a),$$

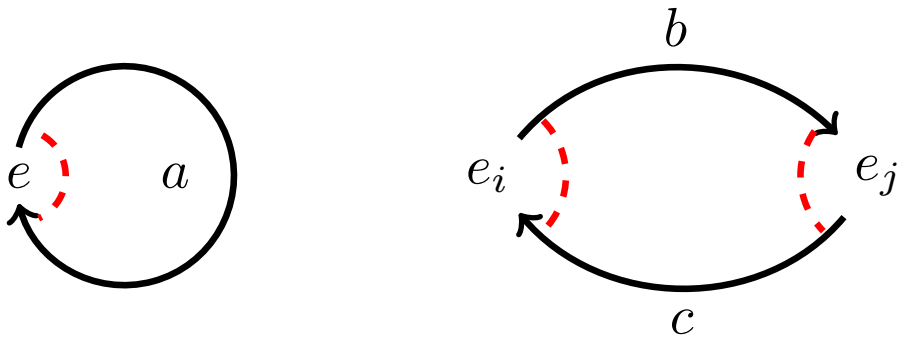
$$d_1(a, \gamma^*) = \sum_{p = \theta a \nu} (p, \theta \gamma^* \nu).$$

# $HH_1(A)^*$

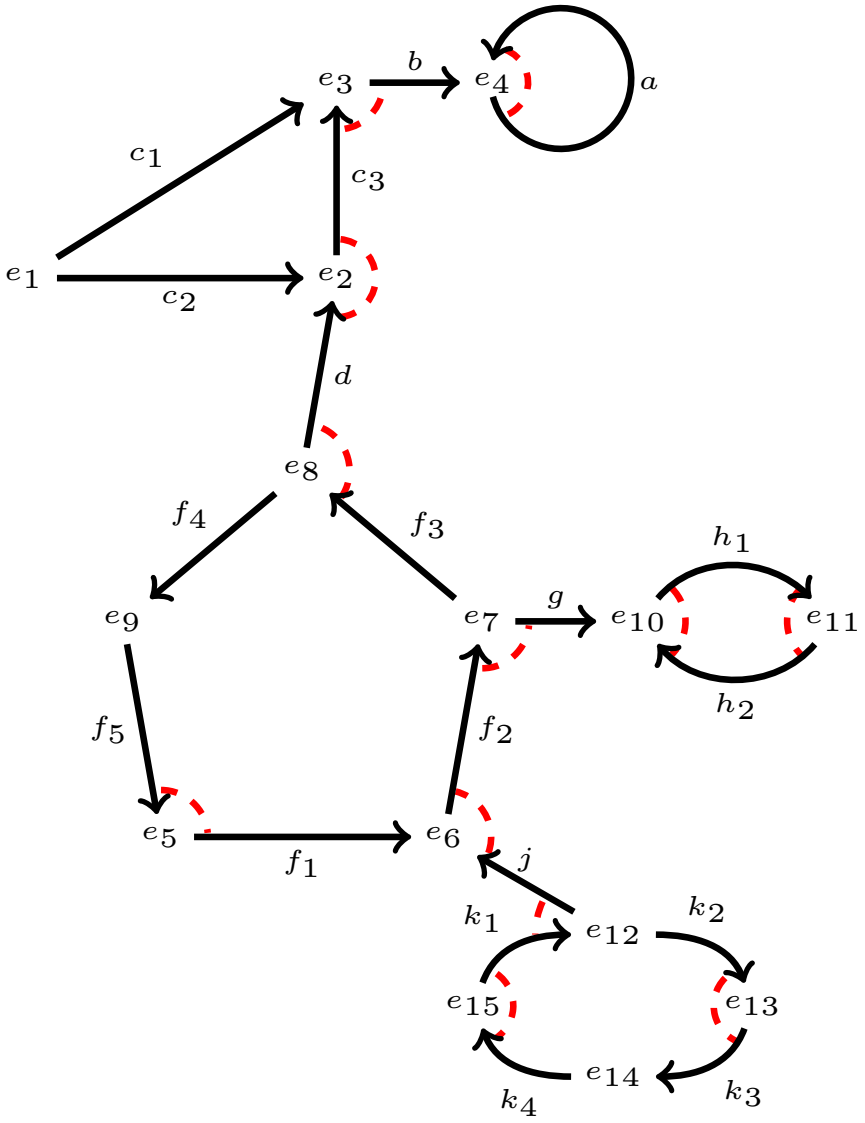
## Theorem (Schroll-Solotar-C)

*Let $A$ be a finite dimensional gentle algebra. The classes of the following elements form a basis of $HH_1(A)^*$:*

- *The pairs $(a, e^*) \in Q_1 \odot Q_0^*$,*
- *The pairs $(b, c^*) - (c, b^*) \in Q_1 \odot Q_1^*$ where $bc = 0 = cb$.*



# $HH_1(A)^*$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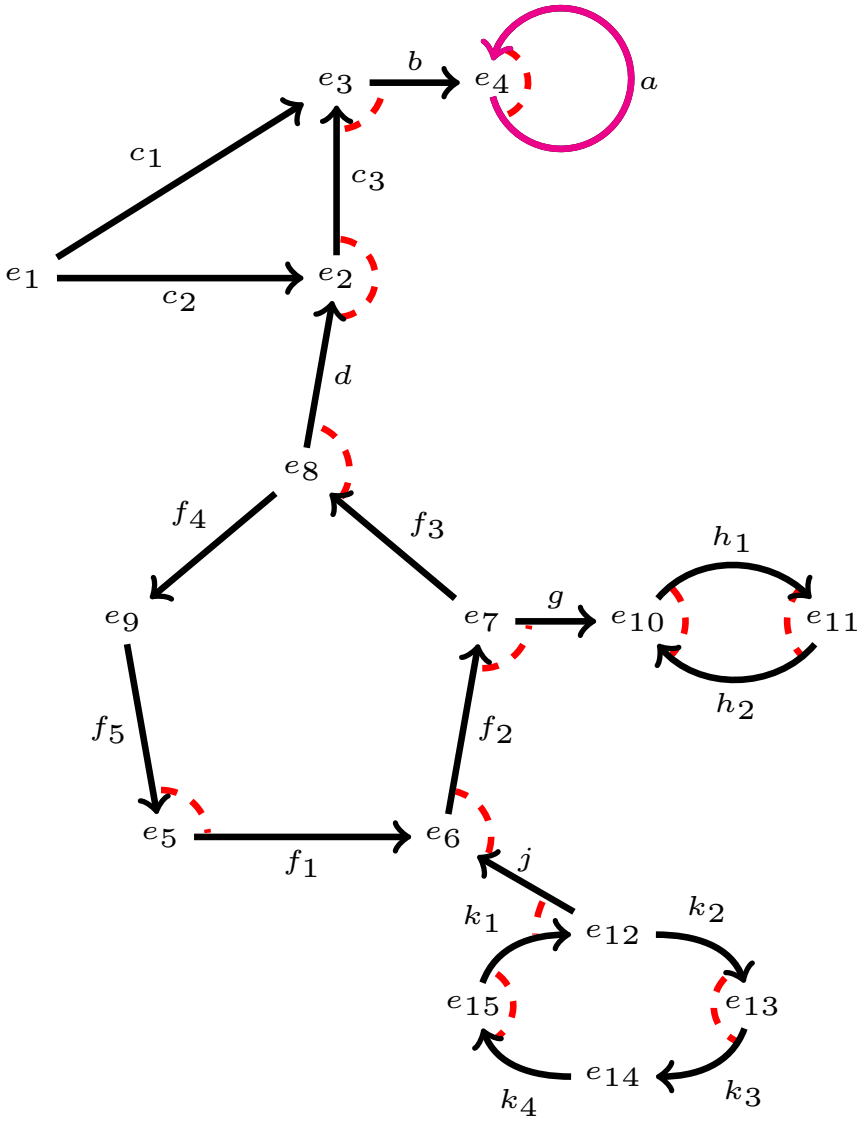
# $HH_1(A)^*$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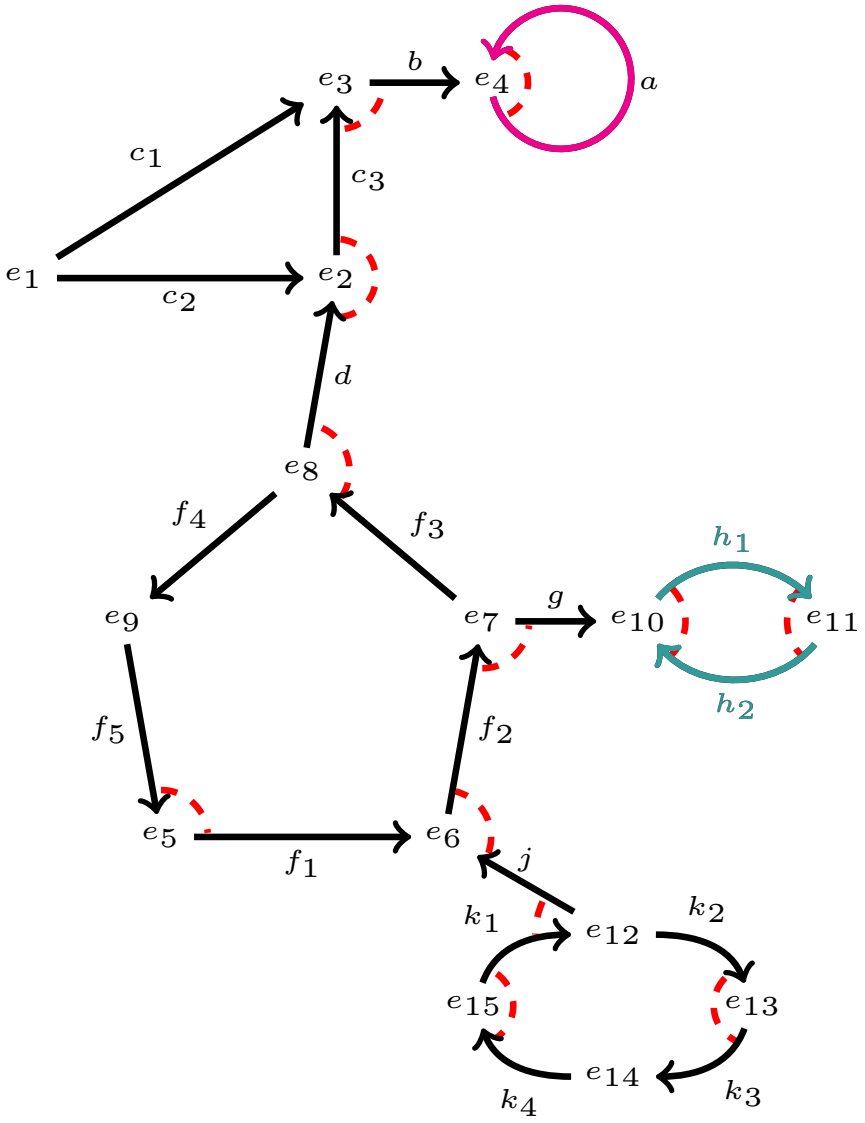
# $HH_1(A)^*$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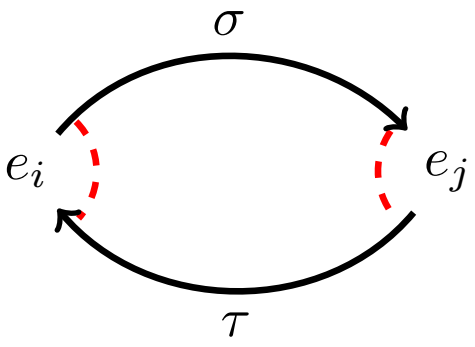
# $Alt_A(DA)$

$$Alt_A(DA) = \{\varphi \in Hom_{A-A}(DA, A) : f\varphi(g) + \varphi(f)g = 0\}.$$

# $Alt_A(DA)$:

## Theorem (Schroll-Solotar-C)

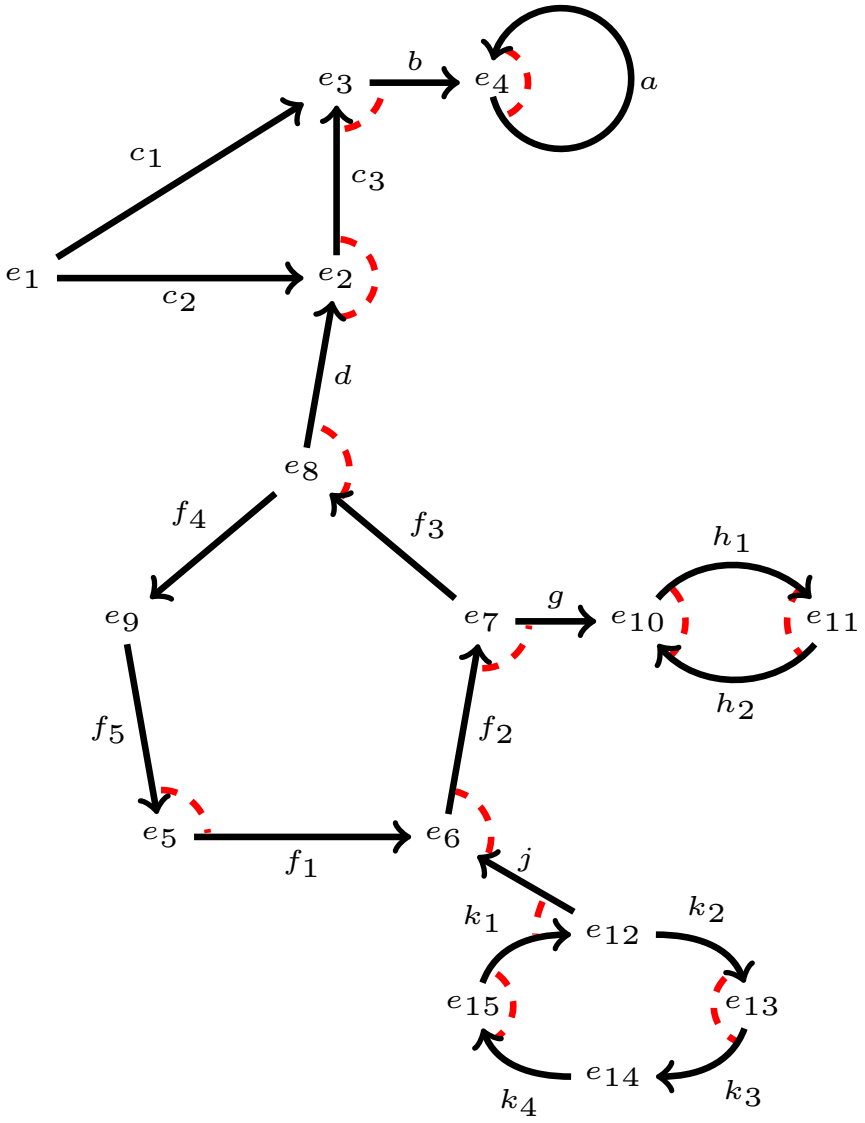
*Let $A$ be a finite dimensional gentle algebra. A basis of $Alt_A(DA)$ is given by the set of elements $(\sigma^*, \tau) - (\tau^*, \sigma)$ where $\sigma\tau = 0 = \tau\sigma$, $val(s(\sigma)) = 2 = val(s(\tau))$.*




## Remark

The cycle $\tau\sigma$ is not necessarily simple.

# $Alt_A(DA)$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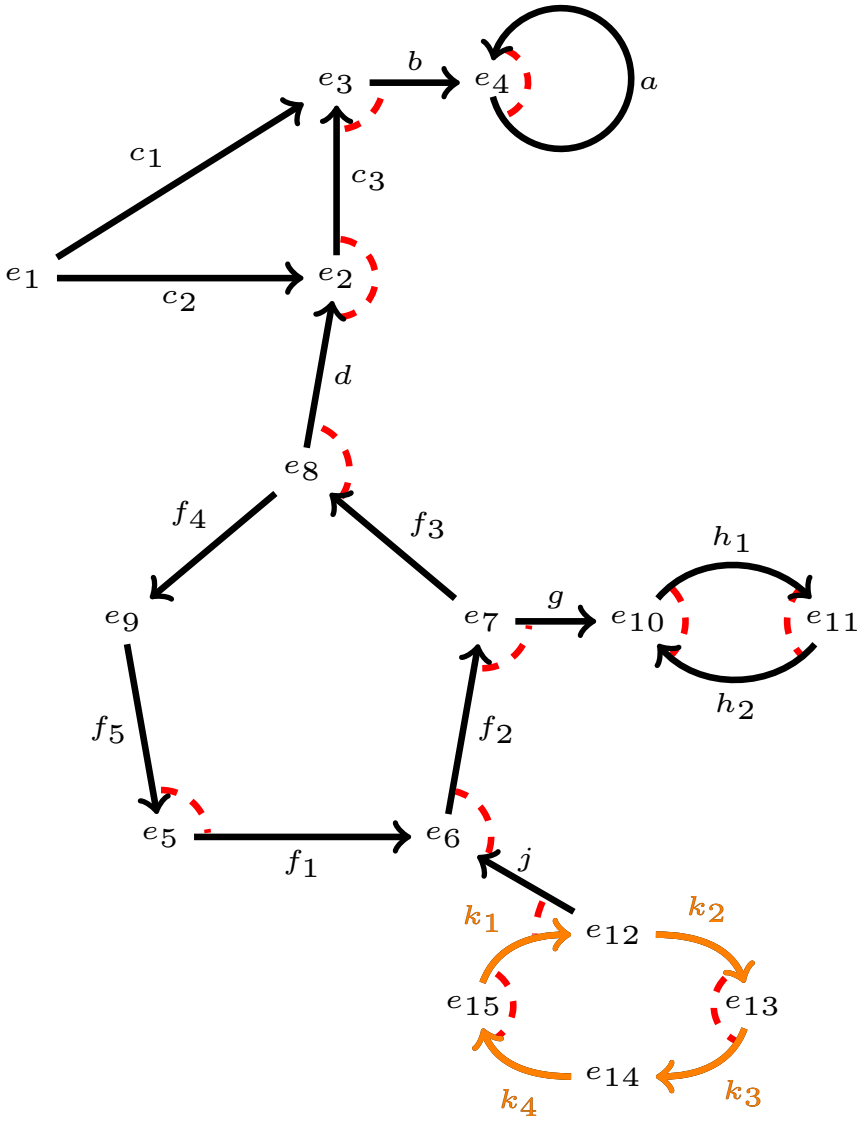
# $Alt_A(DA)$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# The Lie algebra structure of $HH^1(TA)$

**Theorem (Strametz, 2002)**

*Let $A$ be a finite dimensional monomial algebra.*

1. $\ker d^1 / \operatorname{Im} d^0$ *is a Lie algebra with the bracket*

$$[(a,\alpha),(b,\beta)] = (b,\beta^{(a,\alpha)}) - (a,\alpha^{(b,\beta)}),$$

   *where* $(a,\alpha)$, $(b,\beta) \in Q_1||B$.

2. $HH^1(A)$ *and* $\ker d^1 / \operatorname{Im} d^0$ *are isomorphic Lie algebras.*

# The Lie algebra structure of $HH^1(TA)$

## Theorem (Strametz, 2002)

*Let $A$ be a finite dimensional monomial algebra.*

- *The image of $HH^1(A)$ with the previous bracket is a Lie subalgebra of $HH^1(TA)$.*
- *The factors $Z(A)$, $H^1(A, DA)$ and $Alt_A(DA)$ are abelian Lie subalgebras of $HH^1(TA)$.*

| | | | |
|---|---|---|---|
| $[-,-]$ | $Alt_A(DA)$ | $HH_1(A)^*$ | $HH^1(A)$ |
| $Z(A)$ | $Alt_A(DA)$ | $HH_1(A)^*$ | $Z(A)$ |
| $HH^1(A)$ | $Alt_A(DA)$ | $HH_1(A)^*$ | |
| $HH_1(A)^*$ | $Z(A) \oplus HH^1(A)$ | | |

# Example

Non-zero brackets:

- $[(c_1, c_1), (c_1, c_3c_2)] = -(c_1, c_3c_2)$,
- $[(e_5, f_5f_4f_3f_2f_1), (f_1, f_1)] = -(e_5, f_5f_4f_3f_2f_1)$,
- $\left[\sum_{i=1}^{15}(e_i, e_i), ((k_2k_1)^*, k_4k_3) - ((k_4k_3)^*, k_2k_1)\right] =$ $-((k_2k_1)^*, k_4k_3) + ((k_4k_3)^*, k_2k_1)$,
- $\left[\sum_{i=1}^{15}(e_i, e_i), (a, e_4^*)\right] = (a, e_4^*)$,
- $\left[\sum_{i=1}^{15}(e_i, e_i), (h_1, h_2^*) - (h_2, h_1^*)\right] = (h_1, h_2^*) - (h_2, h_1^*)$,
- $[(k_1, k_1), ((k_2k_1)^*, k_4k_3) - ((k_4k_3)^*, k_2k_1)] =$ $((k_2k_1)^*, k_4k_3) - ((k_4k_3)^*, k_2k_1)$,
- $[(a, a), (a, e_4^*)] = -(a, e_4^*)$,
- $[(h_1, h_1), (h_1, h_2^*) - (h_2, h_1^*)] = -(h_1, h_2^*) + (h_2, h_1^*)$.

## Theorem (Schroll-Solotar-C)

*Let $A$ be a finite dimensional gentle algebra. Then $HH^1(TA)$ is nilpotent if and only if all of the following conditions hold:*

- $Z(A) = k$,
- $HH_1(A)^* = 0$,
- $Alt_A(DA) = 0$,
- *for all $(p, q)$ in the basis of $HH^1(A)$, we have $p = q$.*

## Corollary (Schroll-Solotar-C)

*Let $A$ be a finite dimensional gentle algebra. If $HH^1(TA)$ is nilpotent then $HH^1(TA)$ is an abelian Lie algebra.*

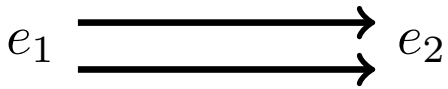
## Theorem (Schroll-Solotar-C)

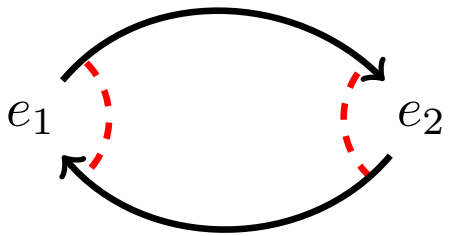
*Let $A$ be a finite dimensional gentle algebra and let $B$ be the Brauer graph algebra isomorphic to $TA$. Suppose $char(k) \neq 2$. Then*

- $HH^1(A)$ *is solvable if and only if $A$ is not the Kronecker algebra.*
- $HH^1(B)$ *is solvable if and only if $B$ is not isomorphic to the trivial extension of the Kronecker algebra.*

## Corollary (Schroll-Solotar-C)

*Let $A$ be a finite dimensional gentle algebra with $char(k) \neq 2$. Then $HH^1(TA)$ is solvable or $HH^1(TA) \cong \mathfrak{gl}(2,k)$. The second case appears when $A$ is given either by the Kronecker algebra or by a Nakayama algebra with $2$ simples and radical square zero.*



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Thanks!